nature portfolio

Peer Review File

Using Genomic Context Informed Genotype Data and Within-model Ancestry Adjustment to Classify Type 2 Diabetes

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This file contains all reviewer reports in order by version, followed by all author rebuttals in order by version.

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)
This manuscript presents a machine learning framework that leverages genotype data and genetic annotations with convolutional neural networks (CNNs) to predict disorder risks. It emphasizes the importance of integrating the current genomic context for enhanced model precision. Furthermore, the authors propose an adversarial ancestry task to control for confounds related to ancestry without affecting model performance on GWAS data. The paper is interesting in using a novel within-model adjustment of ancestry to improve the identification of genomic variants contributing to risk for T2D. However, it appears to be only semi-finished. It requires substantial rework for its presentation, additional method baselines and more rigorous evaluations, particularly in methodological clarification, experimental setup, and linguistic precision. I have the following comments:

1. No interpretation of results. The results only consist of the performance numbers and statistical test p-values. There are not sufficient interpretations of what these numbers suggest and what insights the readers are supposed to learn. Sub-sectioning the results will help. Since there are many auxiliary models, I further suggest the authors to move some baseline models as supplementary information, instead of discussing every model's performance in numbers in the main results. This will help the paper more focused on its main contribution.
2. Lack of independent validation. As a method that aims to provide polygenic risk scores (PRS), it is essential to validate the trained model in an independently collected cohort. Currently it appears the authors are using train-validation-test split in UKBB (but even this is not clear in the paper); the authors should include another non-UKBB cohort for validation of their model.
3. Potential method flaws. The authors concluded using convolutional neural network on CID matrix is the most powerful; however, the CNN's architecture is not fully clear in the paper, and whether it's suitable for this CID data remains questionable.
a) a convolution kernel learns 1D patterns across SNPs (columns) in the CID matrix; however, as each SNP is spatially discontinuous, thus any two columns have variable distance on the genome 1D space, a CNN may not introduce the correct inductive bias on CID.
b) the authors set the height of convolution kernel to be identical as the rows; why not directly use a 1D convolutional operation?
c) Is the CNN picking up different signals than the MLPs? The authors can probe this with a wide range of XAI methods.
4. Additional baseline methods. To show the proposed method is useful, it needs to be put in the broad context of PRS methods. How does the authors CNN-based model compare to conventional methods, including LDPred, PLINK, lassosum2, and PRSice?
5. The authors should specify whether the gradient reversal layer involves one-step or multiple-step attacks and whether it functions similarly to the fast gradient sign method for better clarification and appeal to a wider audience with ML knowledge.
6. In the experiments, how does Figure 1f show that the two input sources undergo the same transformations in the shared layers, yet the output of those transformations is different? Why do Figure 1f and Figure 1g show the same content but different labels? The same issue occurs with Figure 1h and 1i. In Figure 2, the red and green bars are suggested to be removed.
7. The paper is readable, and its structure is clear. However, there are many typos and a significant need for further proofreading. For example, in line 114, "improves classification performance relative to …", there was literally still three dots as placeholders.

Reviewer #2

(Remarks to the Author)
This is an interesting paper comparing a new set of models based on CNNs to classify T2D. Beyond simply the use of CNNs, there were two primary ideas that were closely investigated in this manuscript: (1) the use of a context-informed data matrix (CID), and the use of adversarial learning to “unlearn” information about ancestry as modeled via PCs.

I would like to clarify to both the authors and the editor that I am far from being a NN expert, so I am not commenting about the technical aspects of NNs.

The authors found that using adversarial learning to unlearn ancestry (as modeled via PCs) improved classification of T2D. I am a bit skeptical of this in general, because there were other studies that showed that using PCs is actually useful for prediction and classification. For example, see Naret, Olivier, et al. "Improving polygenic prediction with genetically inferred ancestry." Human Genetics and Genomics Advances 3.3 (2022). I wonder if the authors see what they see because of weird patterns/imbalance of ancestry composition between the training and testing dataset. For example, maybe only the test dataset has individuals with substantial African ancestry, or something like that.

Other comments:
- I think the title should explicitly say “T2D”. It is not clear that the same results will apply to other outcomes/phenotypes. I understand that the approach is general, but the CID didn’t improve classification this time (and maybe it never will) and ancestry may be very helpful for classification of other outcomes.

- A lot of the methods and results are not in the paper. They should be there. Like sample selection, report of the number of SNPs used, etc.

- In line 115, there are three dots “relative to … and whether” did the author not complete the sentence? Or something else weird happened?

- Using PRS in logistic regression: it is worth using a better PRS. There are many PRSs on the PGS catalog, the authors should review and choose a strong PRS.

- There are many models described, sometimes it’s to follow. Perhaps not results from all models should be described thoroughly in the results? In line 260, it should be clarified that the $R2< 0$ refers to the PC estimation.

- The difficulty with statements about cofounding and PCs is that PCs may capture real signal. Meaning, they may be correlated with real genetic determinants that are not known as single SNPs, e.g. because the sample used in GWAS did not represent in a specific way a population with genetic effects (e.g. think about diverse populations, or maybe GxE where population of people with specific characteristics that are related to higher genetic effects of certain variants are under-represented). This agrees with results shown, e.g. as described in lines 290-295. So this duality of potential usefulness of PCs and where they may not be useful deserves a careful discussion (meaning, even more than the existing one, which focuses on the CNNs).

- Can the authors comment on whether the models can be used to reveal something about T2D genetic basis (beyond GWAS)?

- lines 326-327: I think the sentence should be rephrased “we trained a model with genomic context information… and without genomic context information...”, I think you’re trying to say with two types of input.

Version 1:

Reviewer comments:

Reviewer #2

(Remarks to the Author)
Thank you for addressing my comments!

Reviewer #3

(Remarks to the Author)
I have been requested by the editor to consider Reviewer #1's comments and provide further judgment. Below are my evaluations:
1. No Interpretation of Results and Lack of Independent Validation
1.1 The rebuttal presents correlations between probit-transformed feature importance p-values from the training and testing subsets as evidence of generalizability. However, this approach primarily reflects the consistency in data distribution between these subsets, rather than providing a robust measure of generalizability or the robustness of identified important SNPs. This interpretation overlooks the critical requirement of validating feature importance across independent datasets or

populations, which is essential to establish the robustness and to minimize dataset-specific biases.
1.2 The rebuttal relies on internal correlations without addressing potential overfitting or dataset-specific biases. Without validation on an independent external dataset, claims of generalizability remain speculative. The observed correlations may reflect shared artifacts or characteristics within the dataset rather than true biological signals. This distinction must be clearly acknowledged. To strengthen the manuscript, the authors should consider incorporating external validation or alternative strategies to ensure robustness and mitigate confounding risks.
Additional Comments
CNN Architecture: I have no further comments regarding the CNN's architecture.
Baseline Methods: I have no additional concerns regarding the inclusion of baseline methods.
Clarification and Figure Illustrations: I have no further issues with the clarifications or figures provided in the manuscript.

Version 2:

Reviewer comments:

Reviewer #3

(Remarks to the Author)
The authors have addressed the concerns with appropriate external validation using independent WTCCC data. The revised generalization analyses and corresponding results are sufficient to support publication.